MD RAKIBUL ISLAM RAIHAN

ML Systems Engineer | LLM Inference & Multimodal AI

2027 New Grad · Available July 2027 · Open to Relocation · Xi'an, China

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Dear Hiring Manager,

I am writing to express my interest in engineering and research opportunities at your organization. I am a Software Engineering master's student at Northwestern Polytechnical University (NWPU), graduating in July 2027, with hands-on experience in large language model inference, GPU performance optimization, AI infrastructure, and multimodal generative AI research. I am particularly interested in building AI systems that are not only capable, but also efficient, scalable, and reliable in real-world deployment.

At InfiX.ai, I worked on production LLM inference across NVIDIA A800 and H200 GPU infrastructure, tackling performance challenges involving memory management, distributed execution, and model serving. I reduced long-context time to first token from 55 seconds to 6.7 seconds through a three-tier KV-cache offloading architecture spanning GPU HBM, CPU DRAM, and SSD storage over RDMA. I also increased stable serving concurrency from 192 to 256 through prefill-decode disaggregation and RDMA-based KV transfer. In addition, I modified SGLang's cache and memory-management components, investigated production serving failures, and used performance profiling to identify optimization opportunities across the inference stack. These experiences taught me to approach AI performance as a systems problem, connecting software architecture, memory movement, communication overhead, and GPU computation.

Alongside my engineering work, I conduct research at NWPU on efficient multimodal generative AI. As the first author of a manuscript currently under review at IEEE Transactions on Neural Networks and Learning Systems (TNNLS), I have investigated NVFP4 quantization and custom Triton kernels for audio-guided video diffusion transformers. In my evaluated setting, this work achieved a 4x reduction in memory usage, with optimized kernels delivering up to 3.2x speedup over FP16. My research has strengthened my understanding of low-precision computation, GPU kernel optimization, and the practical trade-offs involved in deploying computationally demanding AI models.

I also bring experience in full-stack software development, distributed computing, and containerized AI infrastructure. Across these projects, I have learned to move between experimentation and implementation: identifying bottlenecks, evaluating potential solutions, measuring performance, and building systems that can be reproduced and maintained. I believe this combination of systems-level engineering and research-driven problem solving would allow me to contribute across a range of AI engineering teams, from inference and infrastructure to GPU optimization and multimodal AI.

I would welcome the opportunity to bring my technical background, analytical approach, and commitment to measurable engineering outcomes to your team. I am eager to contribute to challenging AI systems projects while continuing to grow as an engineer working at the intersection of machine learning and high-performance computing.

Thank you for your time and consideration. I look forward to the opportunity to discuss how my experience and interests align with your organization's needs.

Sincerely,

MD RAKIBUL ISLAM RAIHAN

M.Eng. Software Engineering, NWPU (Expected 2027)

MD RAKIBUL ISLAM RAIHAN

ML 系统工程师 | 大模型推理与多模态 AI

2027 届应届 · 2027 年 7 月到岗 · 可接受异地 · 中国·西安

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尊敬的招聘负责人：

本人写信表达加入贵公司工程与研究岗位的意愿。我是西北工业大学（NWPU）软件工程硕士研究生，预计 2027 年 7 月毕业，在大语言模型推理、GPU 性能优化、AI 基础设施与多模态生成式 AI 研究方面具备实践经验。我尤其关注构建不仅能力强、而且在真实部署中高效、可扩展且可靠的 AI 系统。

在 InfiX.ai，我基于 NVIDIA A800 与 H200 GPU 基础设施从事生产级大模型推理工作，解决涉及内存管理、分布式执行与模型服务的性能挑战。我通过跨越 GPU HBM、CPU DRAM 与 SSD 的三级 KV 缓存卸载架构（基于 RDMA），将长上下文首字延迟由 55 秒降至 6.7 秒；并通过 prefill-decode 分离与基于 RDMA 的 KV 传输，将稳定服务并发由 192 提升至 256。此外，我修改了 SGLang 的缓存与内存管理组件，排查生产服务故障，并使用性能分析定位整个推理栈的优化空间。这些经历让我学会把 AI 性能当作系统问题来处理——将软件架构、内存搬运、通信开销与 GPU 计算联系在一起。

在工程工作之外，我在 NWPU 从事高效多模态生成式 AI 研究。作为目前在 IEEE 神经网络与学习系统汇刊（TNNLS）审稿中的论文第一作者，我研究了面向音频引导视频扩散 Transformer 的 NVFP4 量化与自研 Triton 算子。在我的评测设置中，该工作实现了 4 倍显存降低，优化后的算子相较 FP16 最高带来 3.2 倍加速。这段研究加深了我对低精度计算、GPU 算子优化，以及部署高算力 AI 模型时实际权衡的理解。

我还具备全栈软件开发、分布式计算与容器化 AI 基础设施的经验。在这些项目中，我学会了在实验与实现之间往返：定位瓶颈、评估候选方案、测量性能，并构建可复现、可维护的系统。我认为这种系统级工程与研究驱动的问题求解相结合的能力，使我能够为从推理与基础设施，到 GPU 优化与多模态 AI 的各类 AI 工程团队作出贡献。

我希望能有机会将我的技术背景、分析方法以及对可度量工程成果的坚持带入贵团队。我渴望参与富有挑战性的 AI 系统项目，并在机器学习与高性能计算的交叉领域持续成长为一名工程师。

感谢您抽出时间阅读。期待有机会进一步沟通我的经验与兴趣如何契合贵公司的需求。

此致 敬礼

MD RAKIBUL ISLAM RAIHAN

软件工程硕士，西北工业大学（预计 2027 年毕业）